

Tom Stesco

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PROGRAMMING

EXPERT

Python

PROFICIENT

C++ • SQL • Bash

TOOLS

vLLM • PyTorch

Airflow • Apache Spark

Kafka • Pub/Sub • Protobuf

Docker • Kubernetes • Terraform

GCP • AWS • Azure

GNU/Linux • Git

EDUCATION

ETH ZURICH

MSc INTEGRATED BUILDING

SYSTEMS

Excellence Scholarship &

Opportunity Programme (ESOP)

Oct 2018 | Zurich, Switzerland

UNIVERSITY OF WATERLOO

BASc HONOURS CIVIL ENGINEERING

Merit Scholarship

Jun 2016 | Waterloo, Canada

RESEARCH

AUTOMATIC CONTROL LAB

Oct 2018 | Zurich, Switzerland

MSc thesis: [Humans in the Loop](#)

Model Predictive Control (MPC) using
online learning from human feedback.

COURSEWORK

GRADUATE

Model Predictive Control

Mathematical Optimization

Computational Physics

Building Control and Automation

Computational Fluid Dynamics

Renewable Energy Technologies

Technology and Innovation Management

Innovation Leadership

Social Networks Research

UNDERGRADUATE

Software Engineering Design Project

Building Science

Bioprocess Engineering

Advanced Calculus

Probability and Statistics

Linear Algebra

EXPERIENCE

TENSTORRENT

Apr 2023 – present (3 years) | Toronto, Canada

Senior Staff Field Application Engineer - AI/ML Software

Sep 2024 – present

- [tt-inference-server](#): co-lead of 20+ engineer cross-functional team developing open-source E2E LLM inference serving framework with integrated benchmarking, evaluations, and deployment tooling for Tenstorrent AI accelerators.
- Architected CI/CD and release system for the inference serving software stack, enabling reliable production LLM deployments and automated regression detection. 18 releases completed, 60+ AI models supported (27 LLMs/VLMs) across 9 different hardware configurations.
- Collaborated with customers' engineering teams to validate and optimize LLM implementations, e.g. quantization, latency/throughput tuning, structured outputs, for production deployment of AI applications. For example, agentic document processing with OCR, multi-modal inputs, and function calling.

Staff Field Application Engineer - AI/ML Software

Apr 2023 – Sep 2024

- Prototyped cloud infra for internet-facing deployment of LLM inference servers running vLLM for enterprise customers, using Docker, AWS EKS, ELB, Azure API Gateway, and Terraform.
- [tt-studio](#): Built open-source customer reference AI applications. e.g. using LangChain and self-hosting LLM inference APIs.
- [tt-buda-benchmarks](#): Designed benchmarking and evaluation pipelines for LLM and CNN models to track model performance and accuracy across AI graph compiler releases.

BCG X

May 2022 – Apr 2023 (1 year) | Toronto, Canada

Senior AI/ML Engineer

May 2022 – Apr 2023

- Led production engineering for petabyte-scale pricing optimization models on Azure Databricks Lakehouse. Internal ETL library enhancements, CI/CD and release process integration with secure artifactory, deployment using Databricks Jobs.
- Led MLOps for time-series XGBoost pricing model lifecycle management: ETL, training, validation, and deployment in production, monitoring, on-call. Using Airflow, Apache Spark, and BigQuery in GCP.

ECOBEE

Oct 2018 – May 2022 (3 years, 8 months) | Toronto, Canada

Senior Data Scientist

Jun 2021 – May 2022

- [US12241649B2](#): Invented and implemented edge ML optimal HVAC control system in embedded C++ using server trained state-space models and hybrid Model Predictive Control (MPC) approach to optimize energy cost over forecasted: energy price, weather, occupancy, heat gains. Resulting in proof of concept experiments and granted patent.
- [building-controls-simulator](#): developed open source HVAC controls digital-twin platform that fits parametric models using real data (terabyte scale) and was used to validate control algorithm changes in simulation. Presented at eSim2020 conference.

Data Scientist

Oct 2018 – Jun 2021

- Scaled internal IoT experimentation and telemetry platform from 0 to 150k+ devices running parameterized experiments by automating build, embedded experiment deployment, telemetry, and monitoring. Using C++, Protobuf, Pub/Sub, Big Query, and Python.